

Instalar dhcp

```
root@ServidorXuan:~# apt install isc-dhcp-server
Installing:
  isc-dhcp-server

Installing dependencies:
  isc-dhcp-common  policycoreutils  selinux-utils

Paquetes sugeridos:
  isc-dhcp-server-ldap  ieee-data

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 103
  Download size: 1.832 kB
  Space needed: 8.002 kB / 14,1 GB available
```

practica 1: DHCP con direccionamiento dinámico.

1. servidor sera autoritativo

rango esra pc.1.0.150/16 a la pc.1.0.160/16

gateway: servidor

DNS: servidor

dominoi: xuan.local

periodo de asignacion: 60s

```
GNU nano 8.4 /etc/dhcp/dhcpd.conf
authoritative;
default-lease-time 60;
max-lease-time 7200;
#ddns-update-style none;

subnet 4.0.0.0 netmask 255.0.0.0 {
  range 4.1.150.0 4.1.160.0;
  option routers 4.0.0.1;
  option domain-name "xuan.local";
  option domain-name-servers 4.0.0.1;
}
```

2. inicia cliente linux, pedir ip y ver fichero de servidor

```
GNU nano 8.4 /var/lib/dhcp/dhcpd.leases
cltt 4 2025/12/04 11:40:32;
binding state active;
next binding state free;
rewind binding state free;
hardware ethernet 08:00:27:a7:d7:b6;
uid "\377'\247\327\266\000\001\000\0010\301\276\363\010";
client-hostname "debian13-Xuan";
}
lease 4.1.150.0 {
  starts 4 2025/12/04 11:42:50;
  ends 4 2025/12/04 11:47:50;
  cltt 4 2025/12/04 11:42:50;
  binding state active;
  next binding state free;
  rewind binding state free;
  hardware ethernet 08:00:27:a7:d7:b6;
  uid "\377'\247\327\266\000\001\000\0010\301\276\363\010";
  client-hostname "debian13-Xuan";
}
```

3 una captura de la configuracion de red adquirida por el cliente.

Ifconfig necesita paquete "net-tools"

```
root@debian13-Xuan:~# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 4.1.150.0 netmask 255.0.0.0 broadcast 4.255.255.255
    inet6 fe80::a00:27ff:fea7:d7b6 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:a7:d7:b6 txqueuelen 1000 (Ethernet)
    RX packets 34 bytes 4582 (4.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 50 bytes 6965 (6.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 81 bytes 16770 (16.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 81 bytes 16770 (16.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@debian13-Xuan:~# route -n
Kernel IP routing table
Destination        Gateway            Genmask           Flags Metric Ref    Use Iface
0.0.0.0            4.0.0.1           0.0.0.0           UG    0      0      0 enp0s3
4.0.0.0            0.0.0.0           255.0.0.0         U     0      0      0 enp0s3

root@debian13-Xuan:~# cat /etc/resolv.conf
domain xuan.local
search xuan.local
nameserver 4.0.0.1
root@debian13-Xuan:~#
```

4. desactiva la tarjeta de red obser que nos dice a que servidor le devuelve la IP

```
root@debian13-Xuan:~# ifdown enp0s3
Killed old client process
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/enp0s3/08:00:27:a7:d7:b6
Sending on   LPF/enp0s3/08:00:27:a7:d7:b6
Sending on   Socket/fallback
DHCPRELEASE of 4.1.150.0 on enp0s3 to 4.0.0.1 port 67
```

activala de nuevo y espera que coja IP

```
root@debian13-Xuan:~# ifup enp0s3
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/enp0s3/08:00:27:a7:d7:b6
Sending on   LPF/enp0s3/08:00:27:a7:d7:b6
Sending on   Socket/fallback
DHCPDISCOVER on enp0s3 to 255.255.255.255 port 67 interval 4
DHCPOFFER of 4.1.150.0 from 4.0.0.1
DHCPREQUEST for 4.1.150.0 on enp0s3 to 255.255.255.255 port 67
DHCPACK of 4.1.150.0 from 4.0.0.1
bound to 4.1.150.0 -- renewal in 133 seconds.
```

5, apaga servicio hasta active la tarjeta

```
root@DDNS:~# /etc/init.d/isc-dhcp-server stop
Stopping isc-dhcp-server (via systemctl): isc-dhcp-server.service.
root@DDNS:~#
```

va tener ip hasta no le caduque o reinicia network

```
root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet 4.1.150.0/8 brd 4.255.255.255 scope global dynamic enp0s3
        valid_lft 267sec preferred_lft 267sec
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link dadfailed tentative proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian13-Xuan:~# /etc/init.d/networking restart
Restarting networking (via systemctl): networking.service
^C
root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link dadfailed tentative proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian13-Xuan:~#
```

6, hace en windows

```
root@DDNS:~# /etc/init.d/isc-dhcp-server start
Starting isc-dhcp-server (via systemctl): isc-dhcp-server.service.
root@DDNS:~#
```

en windows

```
C:\Users\usuario>ipconfig

Configuración IP de Windows

Adaptador de Ethernet Ethernet:

    Sufijo DNS específico para la conexión. . : xuan.local
    Vínculo: dirección IPv6 local. . . : fe80::2d5f:b318:f65:c7a1%4
    Dirección IPv4 de configuración automática: 169.254.199.161
    Máscara de subred . . . . . : 255.255.0.0
    Puerta de enlace predeterminada . . . . . :
```

hace ipconfig /renew :

```
C:\Users\usuario>ipconfig /renew

Configuración IP de Windows

Adaptador de Ethernet Ethernet:

    Sufijo DNS específico para la conexión. . . : xuan.local
    Vínculo: dirección IPv6 local. . . . : fe80::2d5f:b318:f65:c7a1%4
    Dirección IPv4. . . . . : 4.1.150.1
    Máscara de subred . . . . . : 255.0.0.0
    Puerta de enlace predeterminada . . . . . : 4.0.0.1
```

Pratica 2: asignacion de una ip fija a un equipo concreto

1. asigne a uno de mis clientes una ip estatica que yo elijo, dentro del rango.

Servidor:

/etc/dhcp/dhcpd.conf

```
host linux1 {
    hardware ethernet 08:00:27:a7:d7:b6;
    fixed-address 4.1.155.155;
}
```

```
root@DDNS:~# /etc/init.d/isc-dhcp-server restart
Restarting isc-dhcp-server (via systemctl): isc-dhcp-server.service.
```

2. Renova ip en cliente:

```
root@debian13-Xuan:~# dhclient enp0s3
root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default q
len 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet 4.1.155.155/8 brd 4.255.255.255 scope global dynamic enp0s3
        valid_lft 298sec preferred_lft 298sec
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link dadfailed tentative proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian13-Xuan:~#
```

3. puede renova con:

windows: ipconfig /renew

linux: dhclient -r [nombreDeInterface] (necesita paquete "isc-dhcp-client")

4. configuracion en cliente

route -n

```
root@debian13-Xuan:~# route -n
Kernel IP routing table
Destination      Gateway          Genmask         Flags Metric Ref    Use Iface
0.0.0.0          4.0.0.1         0.0.0.0         UG    0      0      0 enp0s3
4.0.0.0          0.0.0.0         255.0.0.0       U      0      0      0 enp0s3
```

cat resolv.conf →

```
root@debian13-Xuan:~# cat /etc/resolv.conf
domain xuan.local
search xuan.local
nameserver 4.0.0.1
```

ifconfig:

```
root@debian13-Xuan:~# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 4.1.155.155 netmask 255.0.0.0 broadcast 4.255.255.255
    inet6 fe80::a00:27ff:fea7:d7b6 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:a7:d7:b6 txqueuelen 1000 (Ethernet)
    RX packets 2494 bytes 249000 (243.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1945 bytes 192070 (187.5 KiB)
    TX errors 0 dropped 5 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 229 bytes 51140 (49.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 229 bytes 51140 (49.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

5. cliente: /var/lib/dhcp/dhclient.leases

```
GNU nano 8.4 /var/lib/dhcp/dhclient.leases
rebind 4 2025/12/04 12:18:22;
expire 4 2025/12/04 12:19:00;
lease {
    interface "enp0s3";
    fixed-address 4.1.155.155;
    option subnet-mask 255.0.0.0;
    option routers 4.0.0.1;
    option dhcp-lease-time 300;
    option dhcp-message-type 5;
    option domain-name-servers 4.0.0.1;
    option dhcp-server-identifier 4.0.0.1;
    option domain-name "xuan.local";
    renew 4 2025/12/04 12:18:51;
    rebind 4 2025/12/04 12:20:47;
    expire 4 2025/12/04 12:21:25;
```

6, captura: conf de dhcp

```
GNU nano 8.4 /etc/dhcp/dhcpd.conf
authoritative;
default-lease-time 60;
max-lease-time 7200;
#ddns-update-style none;

subnet 4.0.0.0 netmask 255.0.0.0 {
    range 4.1.150.0 4.1.160.0;
    option routers 4.0.0.1;
    option domain-name "xuan.local";
    option domain-name-servers 4.0.0.1;
}

host linux1 {
    hardware ethernet 08:00:27:a7:d7:b6;
    fixed-address 4.1.155.155;
}
```


log de dhcp.leases:

```
GNU nano 8.4 /var/lib/dhcp/dhcpd.leases
}
lease 4.1.150.1 {
  starts 4 2025/12/04 12:25:02;
  ends 4 2025/12/04 12:30:02;
  cltt 4 2025/12/04 12:25:02;
  binding state active;
  next binding state free;
  rewind binding state free;
  hardware ethernet 08:00:27:af:39:4c;
  uid "\001\010\000\2579L";
  set vendor-class-identifier = "MSFT 5.0";
  client-hostname "DESKTOP-R02VM93";
}
lease 4.1.150.1 {
  starts 4 2025/12/04 12:25:02;
  ends 4 2025/12/04 12:30:02;
  tstp 4 2025/12/04 12:30:02;
  cltt 4 2025/12/04 12:25:02;
  binding state free;
  hardware ethernet 08:00:27:af:39:4c;
  uid "\001\010\000\2579L";
  set vendor-class-identifier = "MSFT 5.0";
}
```

contenido del cliente en /var/lib/dhcp

```
root@debian13-Xuan:~# tree /var/lib/dhcp
/var/lib/dhcp
├── dhclient.enp0s3.leases
└── dhclient.leases

1 directory, 2 files
```

Practica 3: Funcionamiento de dhcp

1 elimina la reserva

```
GNU nano 8.4 /etc/dhcp/dhcpd.conf *
authoritative;
default-lease-time 60;
max-lease-time 7200;
#ddns-update-style none;

subnet 4.0.0.0 netmask 255.0.0.0 {
  range 4.1.150.0 4.1.160.0;
  option routers 4.0.0.1;
  option domain-name "xuan.local";
  option domain-name-servers 4.0.0.1;
}

#host linux1 {
#   hardware ethernet 08:00:27:a7:d7:b6;
#   fixed-address 4.1.155.155;
#}
```

2 renueva linux cliente

```
root@debian13-Xuan:~# dhclient -r | dhclient
Killed old client process
```

```
root@debian13-Xuan:~# dhclient enp0s3
root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred lft forever
    inet6 ::1/128 scope host noprefixroute
        valid lft forever preferred lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet 4.1.150.2/8 brd 4.255.255.255 scope global dynamic enp0s3
        valid lft 299sec preferred lft 299sec
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link dadfailed tentative proto kernel_ll
        valid lft forever preferred lft forever
```

3, mira /var/log/syslog en cliente:

```
dic 04 13:55:12 debian13-Xuan dhclient[3823]: DHCPDISCOVER on enp0s3 to 255.255.255.255 port 67 interval 3
dic 04 13:55:12 debian13-Xuan dhclient[3823]: DHCPPOFFER of 4.1.150.2 from 4.0.0.1
dic 04 13:55:12 debian13-Xuan dhclient[3823]: DHCPREQUEST for 4.1.150.2 on enp0s3 to 255.255.255.255 port 67
dic 04 13:55:12 debian13-Xuan dhclient[3823]: DHCPACK of 4.1.150.2 from 4.0.0.1
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Joining mDNS multicast group on interface enp0s3.IPv4 with address 4.1.150.2.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: New relevant interface enp0s3.IPv4 for mDNS.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Registering new address record for 4.1.150.2 on enp0s3.IPv4.
dic 04 13:55:12 debian13-Xuan dhclient[3823]: bound to 4.1.150.2 -- renewal in 138 seconds.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 4.1.150.2 on enp0s3.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Withdrawing address record for ::1 on lo.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 127.0.0.1 on lo.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Host name conflict, retrying with debian13-Xuan-224
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Registering new address record for fe80::a00:27ff:fea7:d7b6 on enp0s3.*.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Registering new address record for 4.1.150.2 on enp0s3.IPv4.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Registering new address record for ::1 on lo.*.
dic 04 13:55:12 debian13-Xuan avahi-daemon[636]: Registering new address record for 127.0.0.1 on lo.IPv4.
```

en server:

```
dic 04 13:55:11 DDNS dhcpd[2436]: DHCPDISCOVER from 08:00:27:a7:d7:b6 via enp0s3
dic 04 13:55:11 DDNS dhcpd[2436]: DHCPPOFFER on 4.1.150.2 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 04 13:55:11 DDNS dhcpd[2436]: DHCPREQUEST for 4.1.150.2 (4.0.0.1) from 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 04 13:55:11 DDNS dhcpd[2436]: DHCPACK on 4.1.150.2 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 04 13:55:38 DDNS kernel: 12:53:58.698664 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
dic 04 13:55:38 DDNS kernel: 12:55:38.690472 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
dic 04 13:55:38 DDNS kernel: 12:55:38.690787 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
dic 04 13:55:38 DDNS kernel: 12:55:38.691085 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
dic 04 13:55:38 DDNS kernel: 12:55:38.773783 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
dic 04 13:55:38 DDNS kernel: 12:55:38.774383 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
dic 04 13:56:00 DDNS dhcpd[2436]: reuse lease: Lease age 49 (secs) under 25% threshold, reply with unaltered, existing lease for 4.1.150.2
dic 04 13:56:00 DDNS dhcpd[2436]: DHCPREQUEST for 4.1.150.2 from 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 04 13:56:00 DDNS dhcpd[2436]: DHCPACK on 4.1.150.2 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 04 13:56:05 DDNS dhcpd[2436]: reuse lease: Lease age 54 (secs) under 25% threshold, reply with unaltered, existing lease for 4.1.150.2
dic 04 13:56:05 DDNS dhcpd[2436]: DHCPREQUEST for 4.1.150.2 from 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 04 13:56:05 DDNS dhcpd[2436]: DHCPACK on 4.1.150.2 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
```

4, espera un minuto y mira syslog de cliente.

```
dic 04 13:57:46 debian13-Xuan dhclient[3706]: DHCPREQUEST for 4.1.150.2 on enp0s3 to 4.0.0.1 port 67
dic 04 13:57:46 debian13-Xuan dhclient[3706]: DHCPACK of 4.1.150.2 from 4.0.0.1
dic 04 13:57:46 debian13-Xuan dhclient[3706]: bound to 4.1.150.2 -- renewal in 141 seconds.
dic 04 13:57:49 debian13-Xuan dhclient[2579]: DHCPREQUEST for 4.1.150.2 on enp0s3 to 4.0.0.1 port 67
dic 04 13:57:49 debian13-Xuan dhclient[2579]: DHCPACK of 4.1.150.2 from 4.0.0.1
dic 04 13:57:49 debian13-Xuan dhclient[2579]: bound to 4.1.150.2 -- renewal in 117 seconds.
dic 04 13:57:51 debian13-Xuan dhclient[2556]: DHCPREQUEST for 4.1.150.2 on enp0s3 to 4.0.0.1 port 67
dic 04 13:57:51 debian13-Xuan dhclient[2556]: DHCPACK of 4.1.150.2 from 4.0.0.1
dic 04 13:57:51 debian13-Xuan dhclient[2556]: bound to 4.1.150.2 -- renewal in 110 seconds.
dic 04 13:57:58 debian13-Xuan dhclient[3734]: DHCPREQUEST for 4.1.150.2 on enp0s3 to 4.0.0.1 port 67
dic 04 13:57:58 debian13-Xuan dhclient[3734]: DHCPACK of 4.1.150.2 from 4.0.0.1
dic 04 13:57:58 debian13-Xuan dhclient[3734]: bound to 4.1.150.2 -- renewal in 119 seconds.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 4.1.155.155 on enp0s3.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 4.1.150.2 on enp0s3.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Withdrawing address record for ::1 on lo.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 127.0.0.1 on lo.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Host name conflict, retrying with debian13-Xuan-232
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Registering new address record for fe80::a00:27ff:fea7:d7b6 on enp0s3.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Registering new address record for 4.1.155.155 on enp0s3.IPv4.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Registering new address record for 4.1.150.2 on enp0s3.IPv4.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Registering new address record for ::1 on lo.*.
dic 04 13:58:01 debian13-Xuan avahi-daemon[636]: Registering new address record for 127.0.0.1 on lo.IPv4.
```

5. apaga servicio dhcp y mira syslog de cliente.

```
dic 04 14:03:44 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 4.1.150.2 on enp0s3.
dic 04 14:03:44 debian13-Xuan avahi-daemon[636]: Leaving mDNS multicast group on interface enp0s3.IPv4 with address 4.1.150.2.
dic 04 14:03:44 debian13-Xuan avahi-daemon[636]: Interface enp0s3.IPv4 no longer relevant for mDNS.
dic 04 14:03:44 debian13-Xuan dhclient[2463]: DHCPDISCOVER on enp0s3 to 255.255.255.255 port 67 interval 8
dic 04 14:03:44 debian13-Xuan dhclient[3734]: DHCPREQUEST for 4.1.150.2 on enp0s3 to 4.0.0.1 port 67
dic 04 14:03:44 debian13-Xuan dhclient[3734]: send_packet: Network is unreachable
dic 04 14:03:44 debian13-Xuan dhclient[3734]: send_packet: please consult README file regarding broadcast address.
dic 04 14:03:44 debian13-Xuan dhclient[3734]: dhclient.c:3001: Failed to send 300 byte long packet over fallback interface.
dic 04 14:03:45 debian13-Xuan dhclient[3706]: DHCPREQUEST for 4.1.150.2 on enp0s3 to 4.0.0.1 port 67
dic 04 14:03:45 debian13-Xuan dhclient[3706]: send_packet: Network is unreachable
dic 04 14:03:45 debian13-Xuan dhclient[3706]: send_packet: please consult README file regarding broadcast address.
dic 04 14:03:45 debian13-Xuan dhclient[3706]: dhclient.c:3001: Failed to send 300 byte long packet over fallback interface.
```

6, reinicia dhcp y mira syslog de cliente

```
root@DDNS:~# /etc/init.d/isc-dhcp-server start
Starting isc-dhcp-server (via systemctl): isc-dhcp-server.service.
root@DDNS:~#
```

```

root@debian13-Xuan:~# dhclient
root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet 4.1.150.2/8 brd 4.255.255.255 scope global dynamic enp0s3
        valid_lft 299sec preferred_lft 299sec
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link dadfailed tentative proto kernel_ll
        valid_lft forever preferred_lft forever

```

```

dic 04 14:06:49 debian13-Xuan avahi-daemon[636]: Host name conflict, retrying with debian13-Xuan-266
dic 04 14:06:49 debian13-Xuan avahi-daemon[636]: Registering new address record for fe80::a00:27ff:fea7:d7b6 on enp0s3.*.
dic 04 14:06:49 debian13-Xuan avahi-daemon[636]: Registering new address record for 4.1.150.2 on enp0s3.IPv4.
dic 04 14:06:49 debian13-Xuan avahi-daemon[636]: Registering new address record for ::1 on lo.*.
dic 04 14:06:49 debian13-Xuan avahi-daemon[636]: Registering new address record for 127.0.0.1 on lo.IPv4.
dic 04 14:07:09 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 4.1.150.2 on enp0s3.
dic 04 14:07:09 debian13-Xuan avahi-daemon[636]: Withdrawing address record for ::1 on lo.
dic 04 14:07:09 debian13-Xuan avahi-daemon[636]: Withdrawing address record for 127.0.0.1 on lo.

```

7. desconecta el servidor dhcp y activa “avahi-autoipd” en interface enp0s3

```

root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian13-Xuan:~# avahi-autoipd enp0s3
Found user 'avahi-autoipd' (UID 110) and group 'avahi-autoipd' (GID 114).
Successfully called chroot().
Successfully dropped root privileges.
Starting with address 169.254.12.145
Callout BIND, address 169.254.12.145 on interface enp0s3
Successfully claimed IP address 169.254.12.145

```

```

root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet 169.254.12.145/16 brd 169.254.255.255 scope link enp0s3:avahi
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

va dar ip de rango 169,254,0,0/16

en fichero de log

```
dic 11 12:54:46 debian13-Xuan avahi-daemon[655]: Withdrawing address record for 169.254.12.145 on enp0s3.
dic 11 12:54:46 debian13-Xuan avahi-daemon[655]: Leaving mDNS multicast group on interface enp0s3.IPv4 with address 169.254.12.145.
dic 11 12:54:46 debian13-Xuan avahi-daemon[655]: Interface enp0s3.IPv4 no longer relevant for mDNS.
dic 11 12:54:46 debian13-Xuan avahi-daemon[655]: Joining mDNS multicast group on interface enp0s3.IPv4 with address 169.254.12.145.
dic 11 12:54:46 debian13-Xuan avahi-daemon[655]: New relevant interface enp0s3.IPv4 for mDNS.
dic 11 12:54:46 debian13-Xuan avahi-daemon[655]: Registering new address record for 169.254.12.145 on enp0s3.IPv4.
```

8, inicia de nuevo el servicio dhcp e nel servidor. Ver lo que ocurre en el cliente.
vuelve tener ip de rango 4,0,0,0/8 de servidor dhcp

```
root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet 4.1.155.155/8 brd 4.255.255.255 scope global dynamic enp0s3
        valid_lft 114sec preferred_lft 114sec
    inet6 fe80::a00:27ff:fea7:d7b6/64 scope link dadfailed tentative proto kernel_ll
```

9,indicas lo que ocurre cuando cambias el rango de Ips asignables. Observa el fichero de log
/var/log/syslog del cliente y del servidor mientras ocurre.

Cuando el cliente renova su ip, el servidor va mirar el ip de cliente está en su rango. Si está fuera de rango. Va dar otro ip que está entre el rango. Y el cliente va cambiar ip que le da el servidor.

Ahora empieza praticar en Maquina Virtual:

Cambiar rango de ip a 4,20,150,0-4,20,160,0 de servidor

```
GNU nano 8.4 /etc/dhcp/dhcpd.conf *
authoritative;
default-lease-time 60;
max-lease-time 120;
#ddns-update-style none;

subnet 4.0.0.0 netmask 255.0.0.0 {
    range 4.20.150.0 4.20.160.0;
    option routers 4.0.0.1;
    option domain-name "xuan.local";
    option domain-name-servers 4.0.0.1;
}
```

reinicia servicio de dhcp de servidor y mira el syslog y captura:

```
root@DDNS:~# /etc/init.d/isc-dhcp-server restart
Restarting isc-dhcp-server (via systemctl): isc-dhcp-server.service.
root@DDNS:~# journalctl -xe
```

reinicia networking de cliente y mira syslog y captura:

```
root@debian13-Xuan:~# /etc/init.d/networking restart
Restarting networking (via systemctl): networking.service.
root@debian13-Xuan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:d7:b6 brd ff:ff:ff:ff:ff:ff
    altname enx080027a7d7b6
    inet 4.1.155.155/8 brd 4.255.255.255 scope global dynamic enp0s3
```

captura de syslog de cliente

```
dic 11 13:04:22 debian13-Xuan ifup[3170]: DHCPDISCOVER on enp0s3 to 255.255.255.255 port 67 interval 10
dic 11 13:04:23 debian13-Xuan dhclient[3170]: DHCPOFFER of 4.20.150.0 from 4.0.0.1
dic 11 13:04:23 debian13-Xuan ifup[3170]: DHCPOFFER of 4.20.150.0 from 4.0.0.1
dic 11 13:04:23 debian13-Xuan ifup[3170]: DHCPREQUEST for 4.20.150.0 on enp0s3 to 255.255.255.255 port 67
dic 11 13:04:23 debian13-Xuan dhclient[3170]: DHCPREQUEST for 4.20.150.0 on enp0s3 to 255.255.255.255 port 67
dic 11 13:04:23 debian13-Xuan dhclient[3170]: DHCPACK of 4.20.150.0 from 4.0.0.1
dic 11 13:04:23 debian13-Xuan ifup[3170]: DHCPACK of 4.20.150.0 from 4.0.0.1
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Joining mDNS multicast group on interface enp0s3.IPv4 with address 4.20
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: New relevant interface enp0s3.IPv4 for mDNS.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Registering new address record for 4.20.150.0 on enp0s3.IPv4.
dic 11 13:04:23 debian13-Xuan dhclient[3170]: bound to 4.20.150.0 -- renewal in 54 seconds.
dic 11 13:04:23 debian13-Xuan ifup[3170]: bound to 4.20.150.0 -- renewal in 54 seconds.
dic 11 13:04:23 debian13-Xuan systemd[1]: Finished networking.service - Raise network interfaces.
Subject: A start job for unit networking.service has finished successfully
Defined-By: systemd
Support: https://www.debian.org/support

A start job for unit networking.service has finished successfully.

The job identifier is 1579.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Withdrawing address record for 4.20.150.0 on enp0s3.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Withdrawing address record for ::1 on lo.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Withdrawing address record for 127.0.0.1 on lo.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Registering new address record for fe80::a00:27ff:fea7:d7b6 on enp0s3.*
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Registering new address record for 4.20.150.0 on enp0s3.IPv4.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Registering new address record for ::1 on lo.*.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Server startup complete. Host name is debian13-Xuan-51.local. Local serv
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Withdrawing address record for fe80::a00:27ff:fea7:d7b6 on enp0s3.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Withdrawing address record for 4.20.150.0 on enp0s3.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Withdrawing address record for ::1 on lo.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Host name conflict, retrying with debian13-Xuan-52
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Registering new address record for fe80::a00:27ff:fea7:d7b6 on enp0s3.*
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Registering new address record for 4.20.150.0 on enp0s3.IPv4.
dic 11 13:04:23 debian13-Xuan avahi-daemon[655]: Registering new address record for ::1 on lo.*.
```

captura de syslog de servidor

```
dic 11 13:04:07 DDNS isc-dhcp-server[2450]: Launching IPv4 server only.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.150.0: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.150.1: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.150.2: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Lease 4.1.155.155: no subnet.
dic 11 13:04:07 DDNS dhcpd[2463]: Wrote 0 leases to leases file.
dic 11 13:04:07 DDNS kernel: audit: type=1400 audit(1765454647.929:127): apparmor="DENIED" operation="capable" c
dic 11 13:04:07 DDNS dhcpd[2463]: Server starting service.
dic 11 13:04:09 DDNS isc-dhcp-server[2450]: Starting ISC DHCPv4 server: dhcpd.
dic 11 13:04:09 DDNS systemd[1]: Started isc-dhcp-server.service - LSB: DHCP server.
Subject: A start job for unit isc-dhcp-server.service has finished successfully
Defined-By: systemd
Support: https://www.debian.org/support

A start job for unit isc-dhcp-server.service has finished successfully.

The job identifier is 2396.
dic 11 13:04:15 DDNS dhcpd[2463]: DHCPRELEASE of 4.1.155.155 from 08:00:27:a7:d7:b6 via enp0s3 (not found)
dic 11 13:04:21 DDNS dhcpd[2463]: DHCPDISCOVER from 08:00:27:a7:d7:b6 via enp0s3
dic 11 13:04:22 DDNS dhcpd[2463]: DHCPOFFER on 4.20.150.0 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 11 13:04:22 DDNS dhcpd[2463]: DHCPREQUEST for 4.20.150.0 (4.0.0.1) from 08:00:27:a7:d7:b6 (debian13-Xuan) vi
dic 11 13:04:23 DDNS dhcpd[2463]: DHCPACK on 4.20.150.0 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 11 13:05:17 DDNS dhcpd[2463]: DHCPREQUEST for 4.20.150.0 from 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 11 13:05:17 DDNS dhcpd[2463]: DHCPACK on 4.20.150.0 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 11 13:06:18 DDNS dhcpd[2463]: DHCPREQUEST for 4.20.150.0 from 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 11 13:06:18 DDNS dhcpd[2463]: DHCPACK on 4.20.150.0 to 08:00:27:a7:d7:b6 (debian13-Xuan) via enp0s3
dic 11 13:06:20 DDNS kernel: 12:04:29.745725 DrmResizeThread VBoxDRMClient: push screen layout data of 1 display
```